Digital Password Lock System with Attempt Limiting

Sharjeel&Moeez

---

1. Overview:
This project implements a digital password lock circuit using logic gates, a binary counter, and a comparator. The system validates a 4-bit binary password, limits the number of incorrect attempts, and activates visual/audible indicators for valid, invalid, and lockout states.

---

2. Key Components & Their Roles

Component	Function
AND Gate	Confirms complete match (all XORs = 0)
Xor Gate	Compare each bit of user input with stored password
NOT Gate	Inverts logic where needed (e.g., password mismatch signal)
NOR Gates	Ensure clock pulse to counter only under specific conditions
4024 Asynchronous counter	Binary ripple counter; counts wrong attempts
74LS85 Comparator	Comparator; compares attempts vs. maximum allowed
LEDs	Indicate "Valid" (green), "Invalid" (red) states
Buzzer	Activates on lockout

---

3. Operation Flow:

- **Step 1: Password Entry**
  - User enters a 4-bit binary password.
  - XOR gates compare each bit with the preset password.
  - If all bits match, XOR outputs are 0 → AND gate outputs HIGH (match).
- **Step 2: Press Enter**
  - When Enter switch is pressed:
    - If password is correct and not locked out → Green LED lights up.
    - If incorrect and attempts remaining → Red LED lights up, counter increments.
- **Step 3: Invalid Attempts Counter**
  - Counter receives a pulse via NOR gate logic only if:
    - Password is wrong
    - Enter is pressed
    - Max attempts not reached
- **Step 4: Lockout Mechanism**
  - 74LS85 comparator checks if attempts ≥ max (set via switches)
  - If true:
    - Buzzer activates
    - Further inputs are blocked

- **Step 5: Reset**
  - Reset switch clears counter and disables buzzer, allowing retry.

---

5. Logic Control Using NOR Gates:

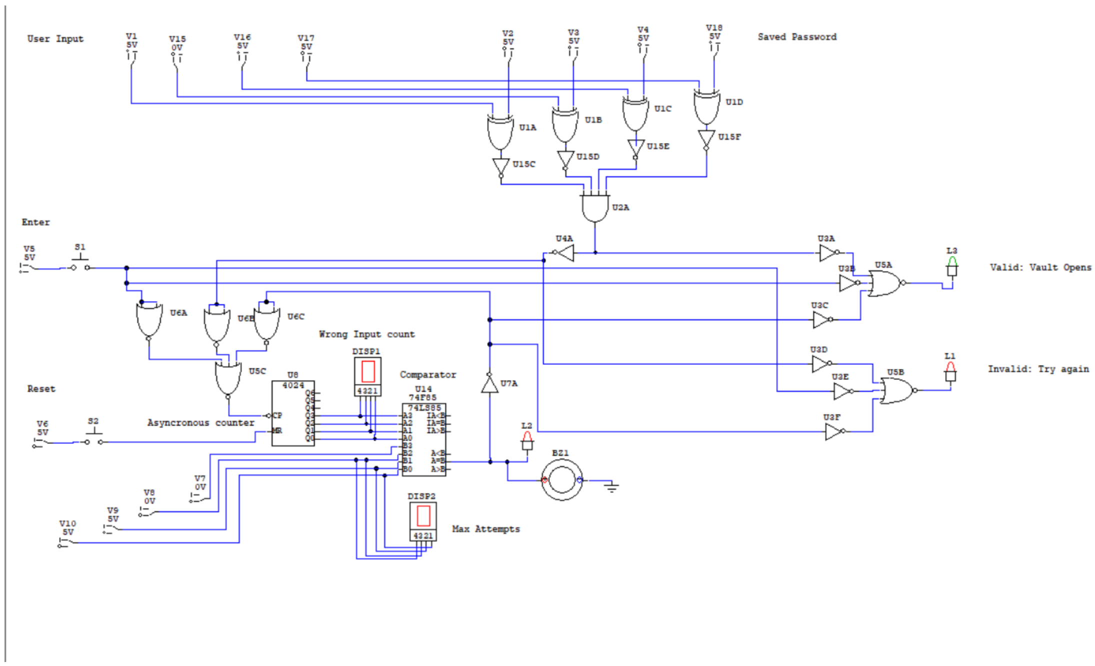
- NOR gate chain ensures that counter only receives clock pulse if:
  - Enter is pressed
  - Password is wrong
  - User is not locked out
- This prevents accidental or repeated counting.

---

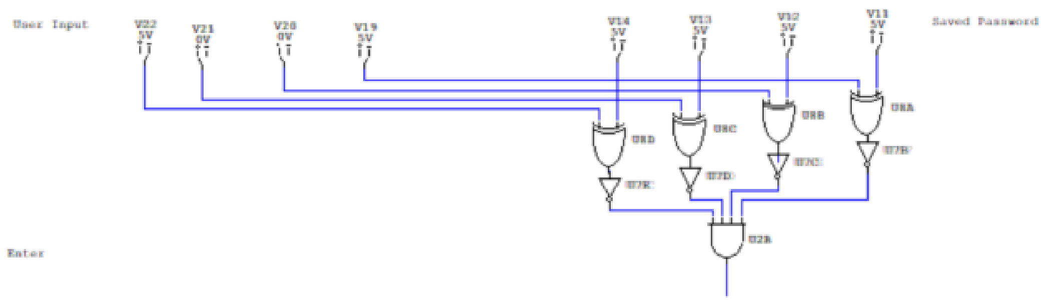
6. Conclusion:

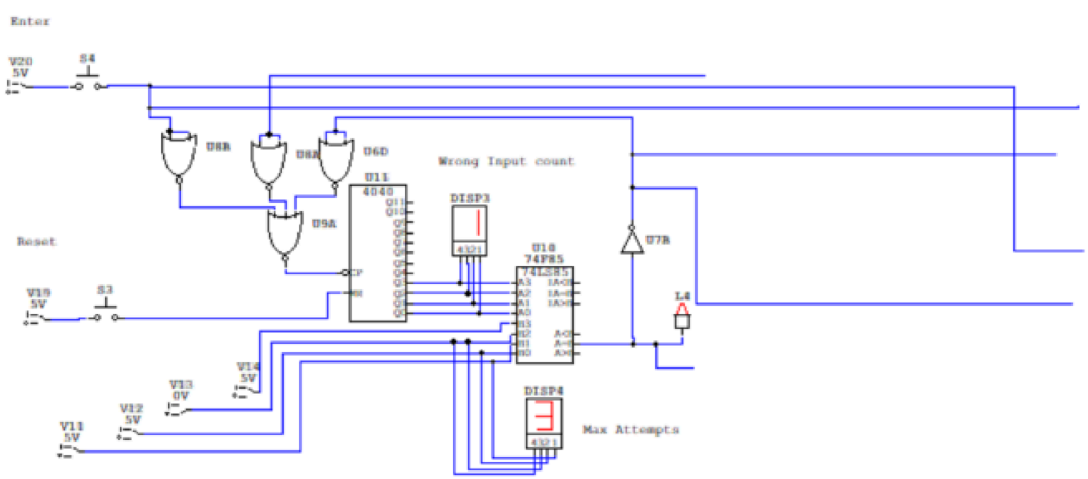
The project demonstrates the integration of combinational and sequential logic for a real-world application: a secure digital lock with limited retry attempts. The system effectively uses logic gates, counter, and comparator to manage validation and security response, making it a solid educational tool for learning digital electronics.

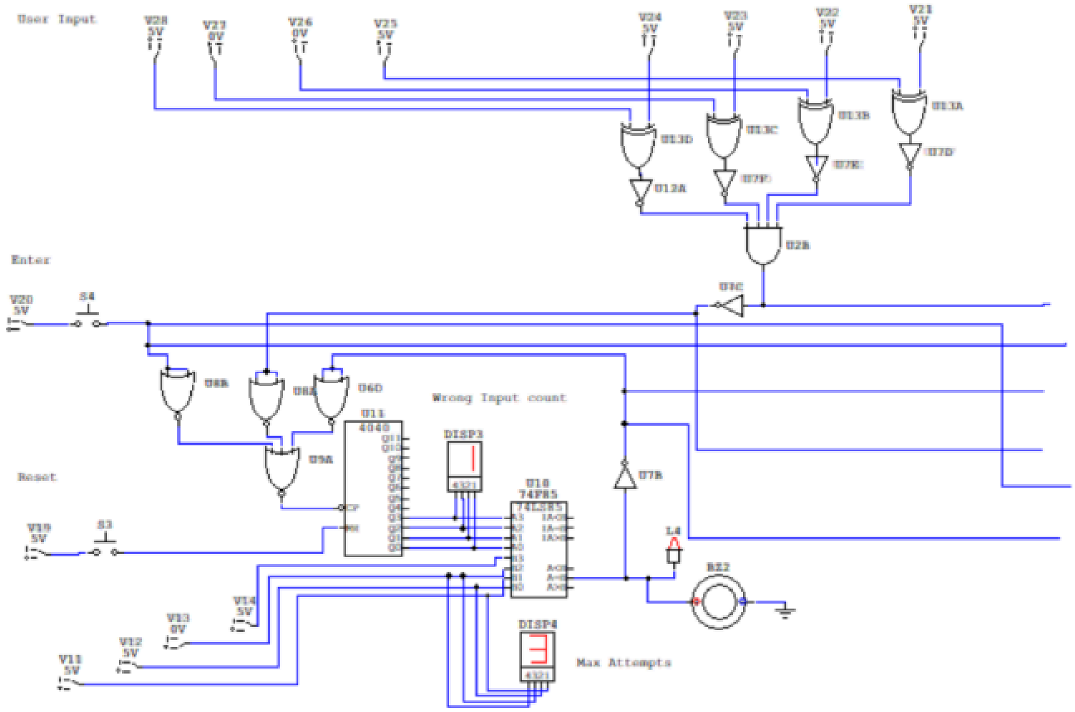
CIrcuit Diagram

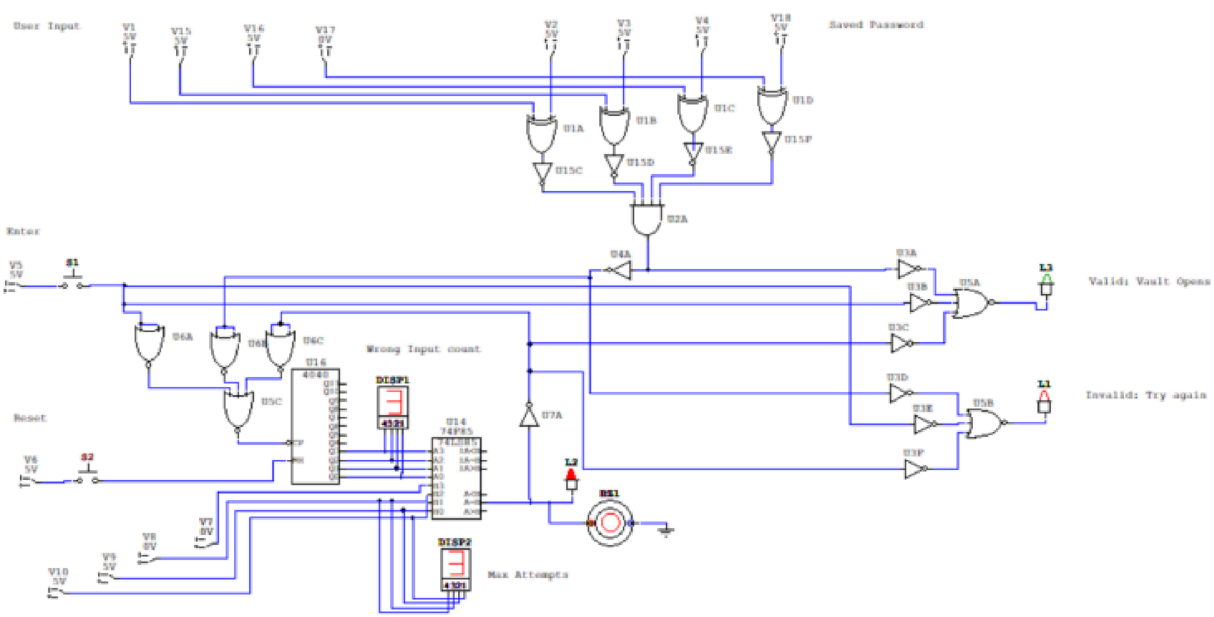


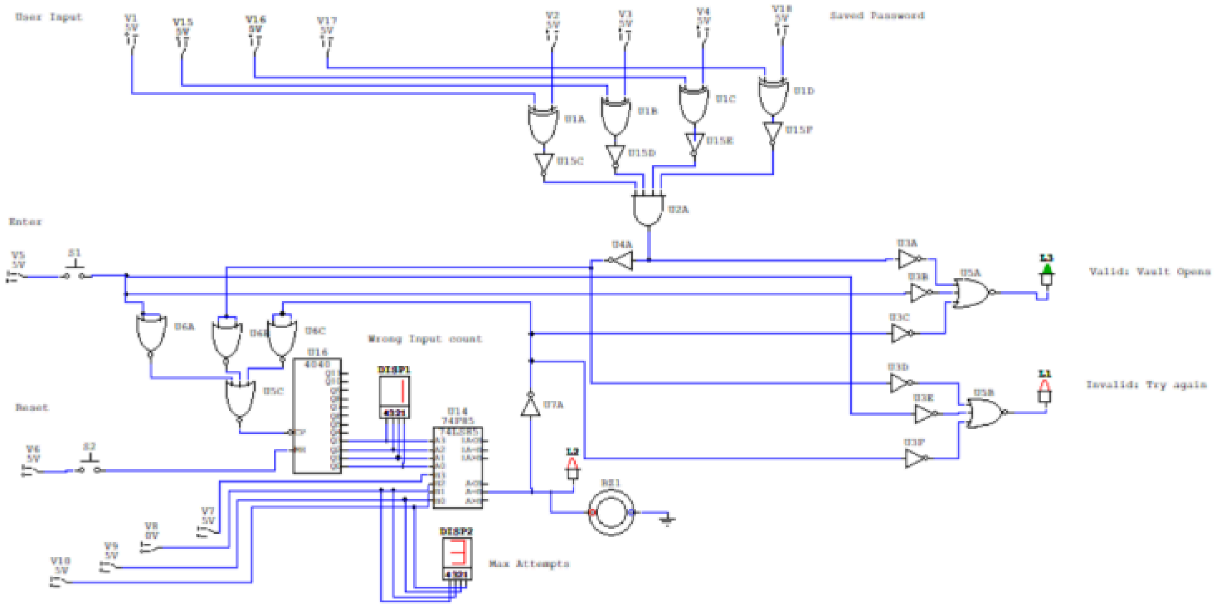
Working

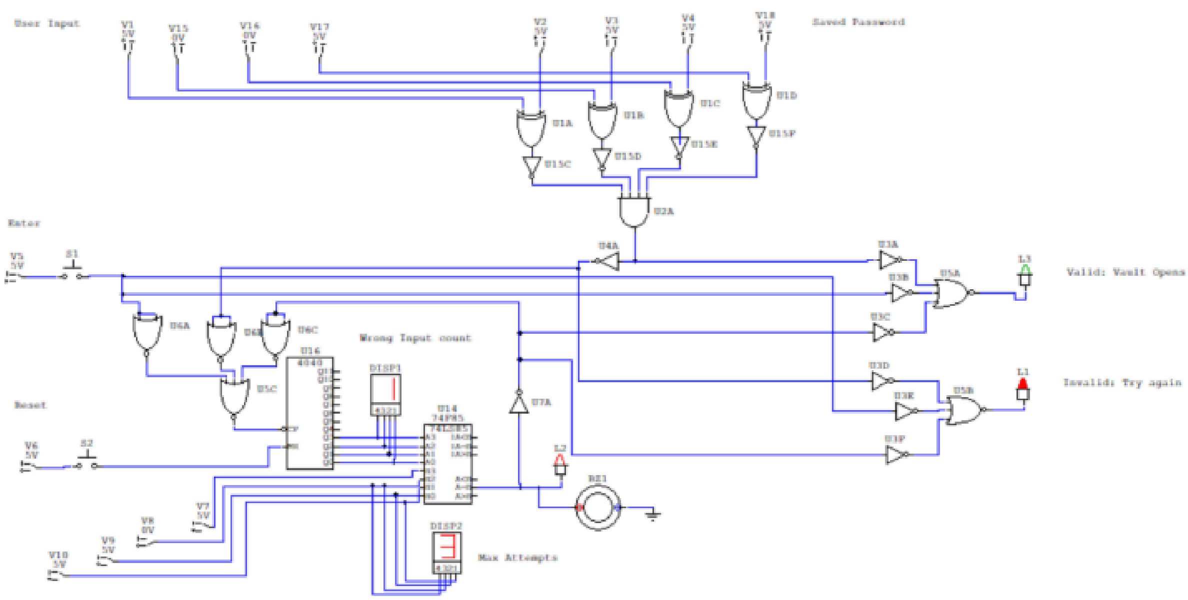

User Input
V22
5V
V21
0V
V20
0V
V19
5V
V14
5V
V13
5V
V12
5V
V11
5V
Saved Password
U8A
U8B
U8C
U8D
U7B
U7C
U7D
U7E
U2B
Enter


Enter
V20
5V
S4
U8B
U6D
Wrong Input count
U11
4040
DISP3
U9A
U10
74F85
U7B
Reset
V15
5V
S3
L4
V14
5V
V13
0V
V12
5V
V11
5V
DISP4
Max Attempts


User Input
Enter
Wrong Input count
Reset
Max Attempts


User Input
Saved Password
Enter
Wrong Input count
Reset
Max Attempts
Valid: Vault Opens
Invalid: Try again


User Input
Saved Password
Enter
Reset
Wrong Input count
Valid: Vault Opens
Invalid: Try again
Max Attempts
DISP1
DISP2
U16
4040
U14
74F85
L1
L2
L3
BZ1
S1
S2


User Input
Saved Password
Enter
Reset
Wrong Input count
Valid: Vault Opens
Invalid: Try again
Max Attempts
DISP1
DISP2
U16
4040
U14
74F85
L1
L2
L3
BZ1
S1
S2